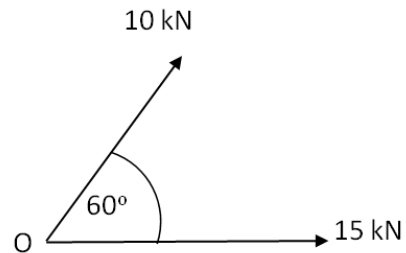


**Mechanics 1**

Tutorial Sheet - 1

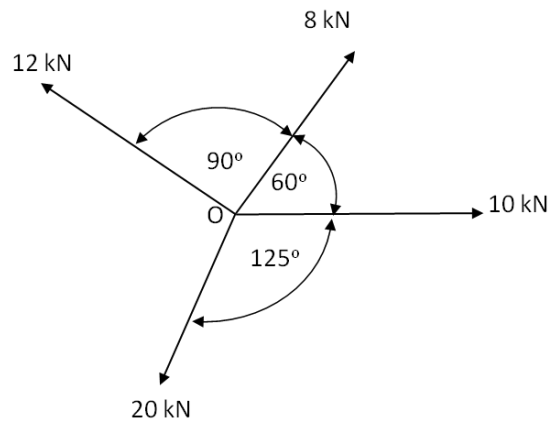
1. Determine the magnitude and inclination of the resultant of the two forces acting at the point O in Fig P1.

**Ans: 21.8 kN, 23.4° to the direction of 15 kN**



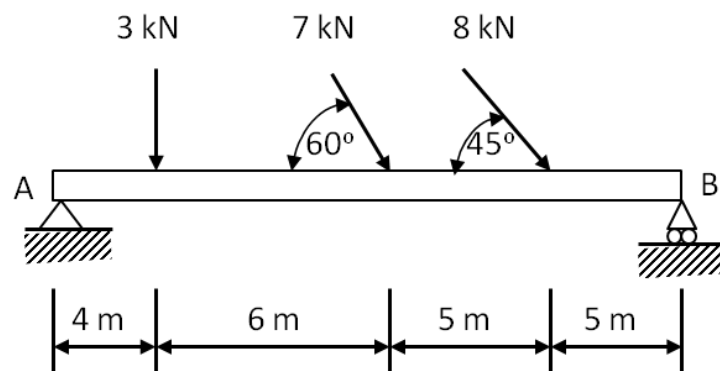
2 Determine the magnitude and inclination of the resultant of the system of concurrent forces acting at the point O in Fig P2.

**Ans: 8.6kN, 23.9° down and to the left**

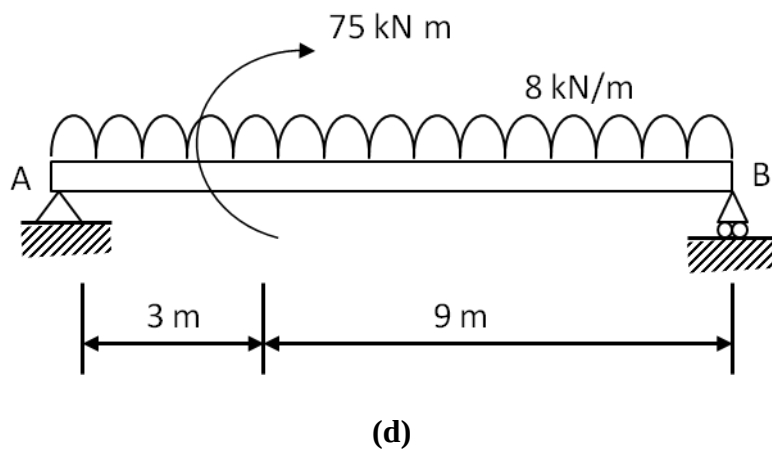
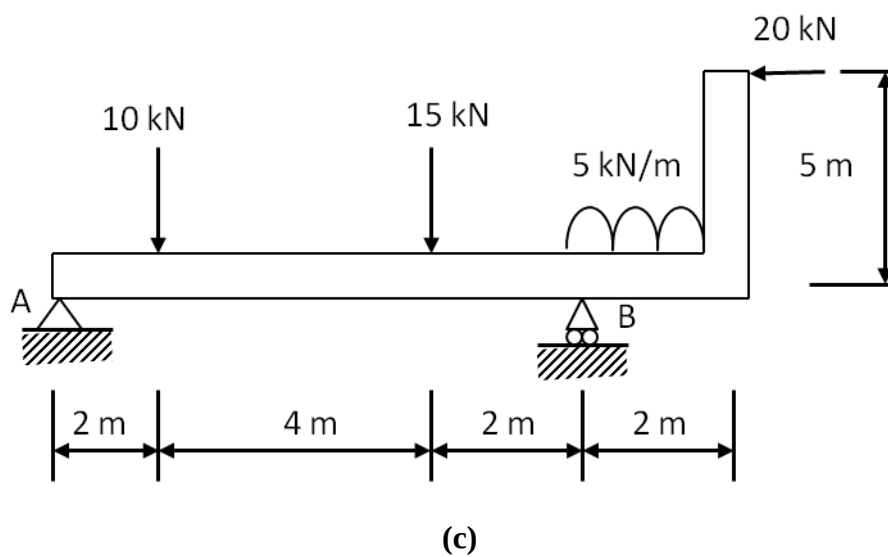
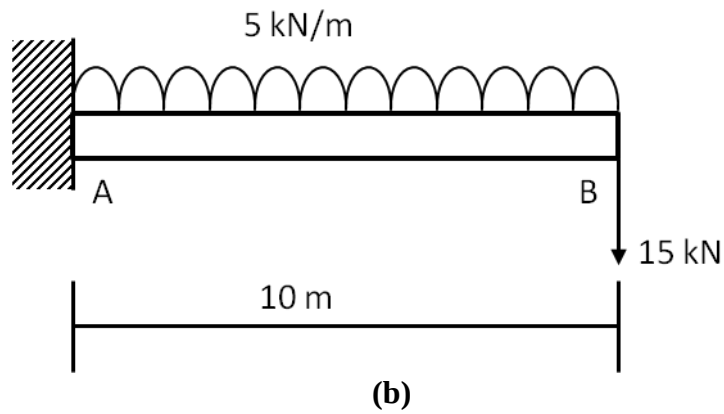


3 Calculate the support reactions in the beams shown in Figs P3(a)-(d)

**Ans:** (a)  $R_{A,H}=9.2$  kN to left,  $R_{A,V}=6.9$  kN upwards,  $R_B=7.9$  kN upwards,  
(b)  $R_A=65$  kN,  $M_A=400$  kN m anticlockwise,  
(b)  $R_{A,H}=20$  kN to right,  $R_{A,V}=22.5$  kN upwards,  $R_B=12.5$  kN upwards,  
(b)  $R_A=41.8$  kN upwards,  $R_B=54.2$  kN upwards.



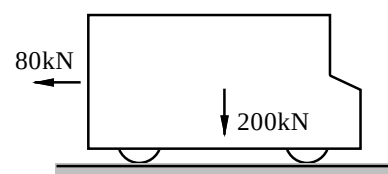
(a)



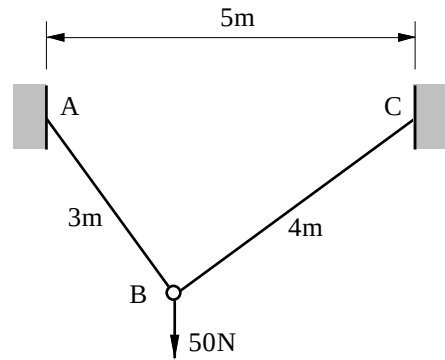
4. A lorry weighing 200kN is subject to a horizontal braking force of 80kN.

What is the resultant force acting on the vehicle, and in what direction does it act?

**Ans: 215.4 kN, 21.8° to vertical**



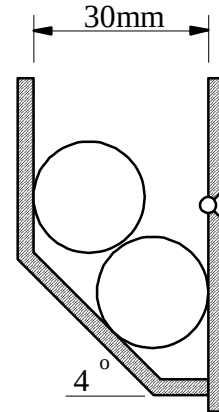
5. An object weighing 50N is supported by two steel wires, AB and BC, of lengths 3m and 4m respectively. The wires are anchored at points A and C, on the same level and 5m apart, as shown in the diagram.



By forming equations for the *horizontal* and *vertical* equilibrium of the object, find the tensile forces in the two wires.

**Ans: 40N, 30N**

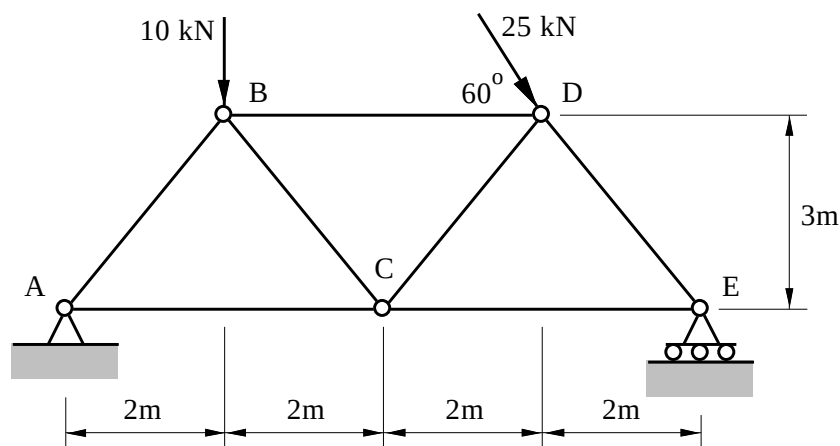
6. Two coins in a slot machine rest against the base of the slot, as shown in the diagram. Each of the coins is 20mm diameter and weighs 0.5N. Friction between the coins and between the coins and the sides of the slot may be neglected. Consequently reactions may be assumed to be perpendicular to the surfaces.



- At what angle to the horizontal does the line joining the centres of the two coins lie?
- By considering equilibrium of the *upper* coin, find the reaction between the two coins.
- By considering equilibrium of the *lower* coin, find the horizontal force acting on the trapdoor at the base.

**Ans:  $60^\circ$ ,  $1/\sqrt{3} = 0.577\text{N}$ ,  $1 + 1/(2\sqrt{3}) = 1.289\text{N}$**

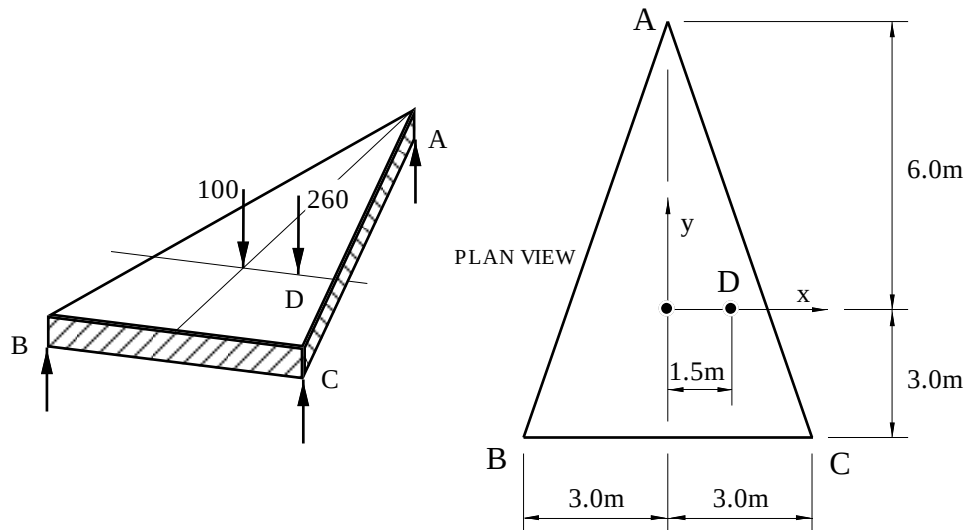
7. The pin-jointed frame shown in the diagram is pinned to a foundation at A and rests on (frictionless) rollers at E. Note that the pin at A is capable of resisting horizontal and vertical forces; the rollers at E can only resist forces normal to their plane.



Form equations for the overall equilibrium of the frame (bearing in mind that the moment of a force is the sum of the moments of its components). Find the external reactions on the frame at A and E.

**Ans:  $V_A = 8.22\text{ kN}$ ,  $H_A = 12.5\text{ kN}$ ,  $V_E = 23.43\text{ kN}$**

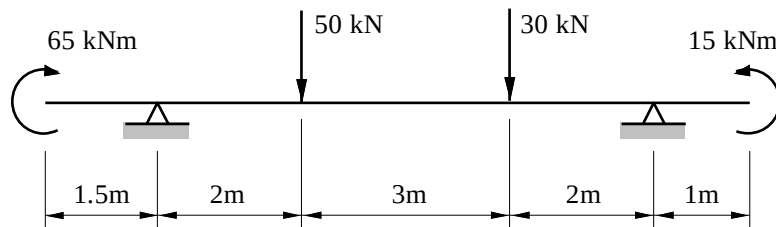
8. A flat, triangular slab is supported by vertical reactions at its corners, as shown in the diagram. The weight of the slab is 100 kN and a vertical load of 260 kN is applied at point D.



Form equations for the overall equilibrium of the slab (forces in *three* directions, moments about *three* axes) and find the external reactions at A, B and C.

**Ans : 120, 55, 185 kN**

9. The horizontal beam shown in the diagram rests on knife-edge supports - which may be assumed to be capable of resisting *vertical* forces only. The beam carries vertical forces of 50 kN and 30 kN and couples of 65 kNm (clockwise) and 15 kNm (anti-clockwise) applied at its ends.



What are the vertical reactions at the supports? (Remember : if a couple has a moment about *one* point, it has exactly the same moment about *any other* point.)

**Ans: 37.14, 42.86 kN**